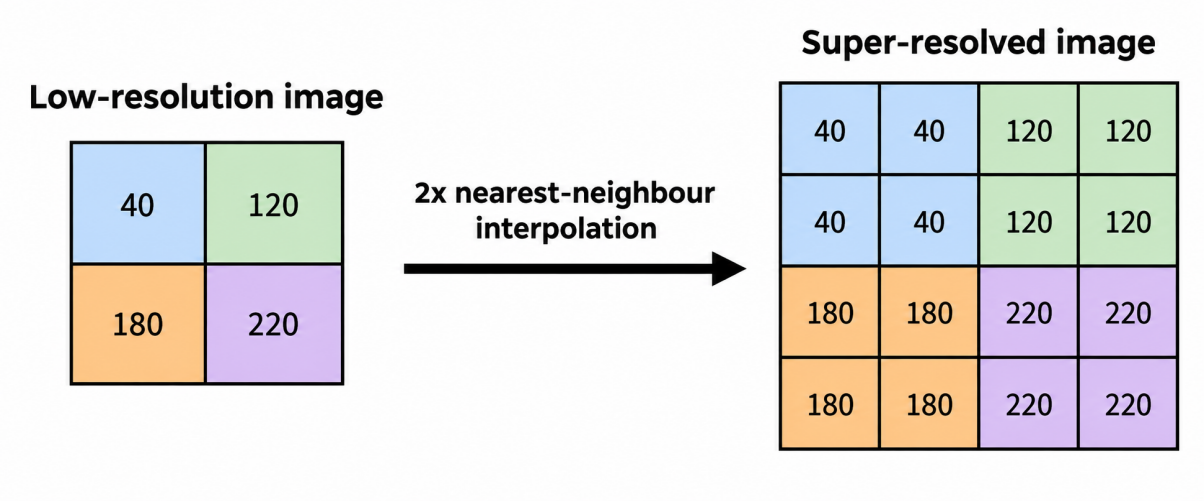
Interpolation-Based Super-Resolution

Prelude: Nearest-Neighbour Interpolation (Unmarked)

Nearest-neighbour interpolation is the simplest super-resolution method. For an enlargement factor of s, each pixel value is copied into an $s \times s$ block. No new pixel intensity is estimated.



For 2× super-resolution, each pixel becomes a 2×2 constant block. For 4× and 8× super-resolution, each pixel becomes a 4×4 and 8×8 constant block, respectively.
Nearest-neighbour interpolation is a zeroth-order technique: it assumes that the intensity remains constant over the region represented by a pixel. It is computationally very fast, but produces blocky or staircase artefacts.
This prelude is unmarked.

Task

Take any four images from the dataset provided in Assignment 1.

1. Convert every selected image to grayscale. The original images are 512×512.
2. Downsample every image to:
   - 64×64
   - 128×128
   - 256×256
3. Super-resolve each downsampled image back to 512×512.
4. Perform interpolation using zeroth-order, first-order, second-order, third-order, etc. methods.
5. For every super-resolved image, compute PSNR and SSIM with respect to the original grayscale 512×512 image.
6. Report the runtime of each interpolation method.

The super-resolution settings are:

Input size	Output size	Enlargement factor
64×64	512×512	$8\times$
128×128	512×512	$4\times$
256×256	512×512	$2\times$

Use the same four images and the same downsampling procedure throughout the study.

Report

The report must include:

- the four selected original grayscale images;
- the three downsampled versions of every image;
- super-resolved results obtained using the interpolation methods;
- PSNR, SSIM, and runtime results;
- clearly labelled tables for comparison;
- a brief discussion of the results.

In the discussion, comment on:

- how quality changes from $2\times$ to $4\times$ and $8\times$ super-resolution;
- the visual differences among the interpolation methods;
- blocking, blur, ringing, overshoot, or staircase artefacts, wherever present;
- the relation between reconstruction quality and runtime;
- whether the observations depend on image content such as edges, smooth regions, or texture.

Deliverables

Submit:

- a report in PDF format;
- Python code or a Python notebook.

The code or notebook must be demonstrated to the TAs.